
Cross-Cohort Replication of Parkinson's Disease Subtypes

Bootstrap Consensus Ensemble with GAMLSS
Normalisation

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Data-Driven Subtypes Are Established

- Data-driven subtypes are well-established within cohorts [Lawton et al. 2018; Fereshtehnejad et al. 2017]
- The MDS Task Force [Marras, Hu et al. 2024] highlights the importance of purpose-driven subtyping for precision medicine

➤ **Subtyping is therefore promising, but its reliability across cohorts remains unclear.**

The Replication Problem

- Cross-cohort replication remains limited
 - Cohort-specific confounds:
 - medication state
 - recruitment
 - assessment instruments
- **Apparent subtype agreement may reflect cohort differences rather than biological structure.**

Positioning This Work

Building on the OPDC subtyping programme (Lawton et al. 2015, 2018)

- Previous OPDC studies identified clinically meaningful PD subtypes within a single cohort

Our contribution addresses a different question

- Rather than discovering new subtypes, we test whether the same patient grouping appears in an independent cohort

Complementary, not competing

- Lawton asks: *"what subtypes exist?"*
- We ask: *"do they replicate across cohort boundaries?"*

➤ **This study is a validation of subtype structure, not a new clustering proposal.**

Experimental Cohort Setup

PPMI: Discovery cohort

- ❖ 794 PD patients
- ❖ Drug-naïve
- ❖ Assessed OFF medication
- ❖ Covariates: age, sex, education

Reference: Prodromal cohort

OPDC: Replication cohort

- ❖ 996 PD patients
- ❖ Treated patients
- ❖ Assessed ON medication
- ❖ Covariates: age, sex, education

Reference: RBD cohort

Primary endpoint:

- Similarity of patient grouping across cohorts (measured by bidirectional ARI).
 - *ARI measures pairwise agreement between cluster assignments (0 = chance | 1 = identical).*

Clustering Features vs Independent Validation Features

Preventing circularity in subtype interpretation

Shared Features

Common clinical representation (PPMI and OPDC biomarkers):

- ❖ UPDRS I (non-motor experiences),
- ❖ UPDRS II (motor ADL),
- ❖ UPDRS III (motor exam),
- ❖ MoCA (cognition),
- ❖ Schwab & England (ADL).

Validation Features

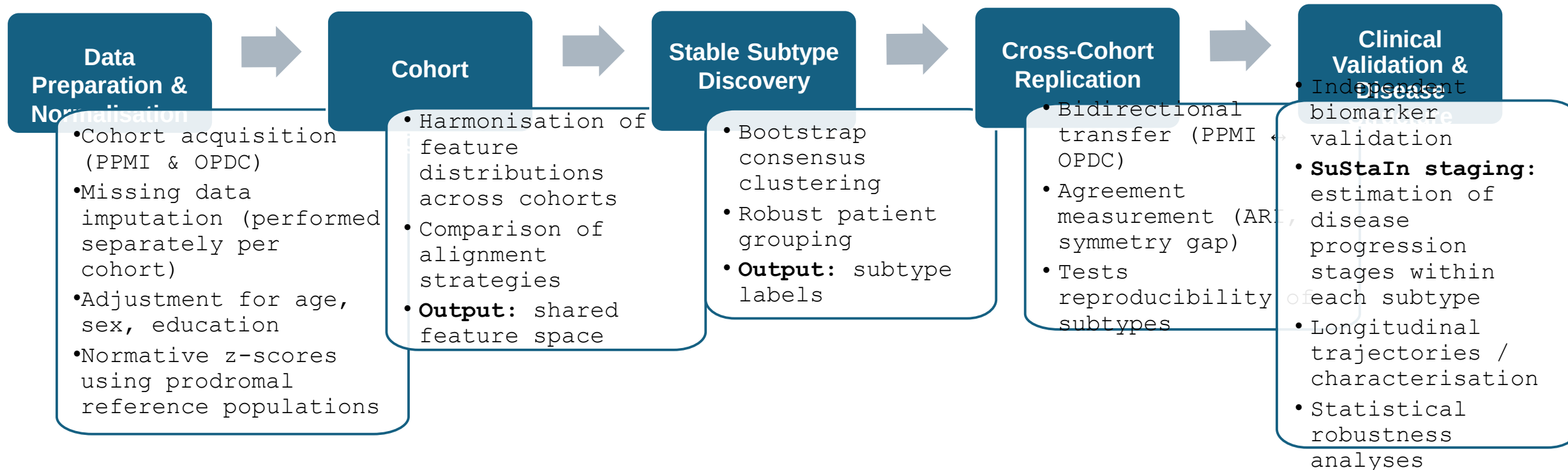
PPMI biomarkers:

- ❖ DaTSCAN dopaminergic imaging,
- ❖ STAI (anxiety),
- ❖ SDMT (processing speed),
- ❖ Grey matter volume (MRI).

OPDC biomarkers:

- ❖ HADS (anxiety/depression),
- ❖ Purdue pegboard (dexterity),
- ❖ Sniffin' Sticks (olfaction),
- ❖ ESS (sleepiness),
- ❖ BDI (Beck Depression Inventory),
- ❖ RBD screening.

Study Workflow



Clustering & Replication

Model Selection Across Harmonisation

Selecting a Reliable Cross-Cohort Clustering:

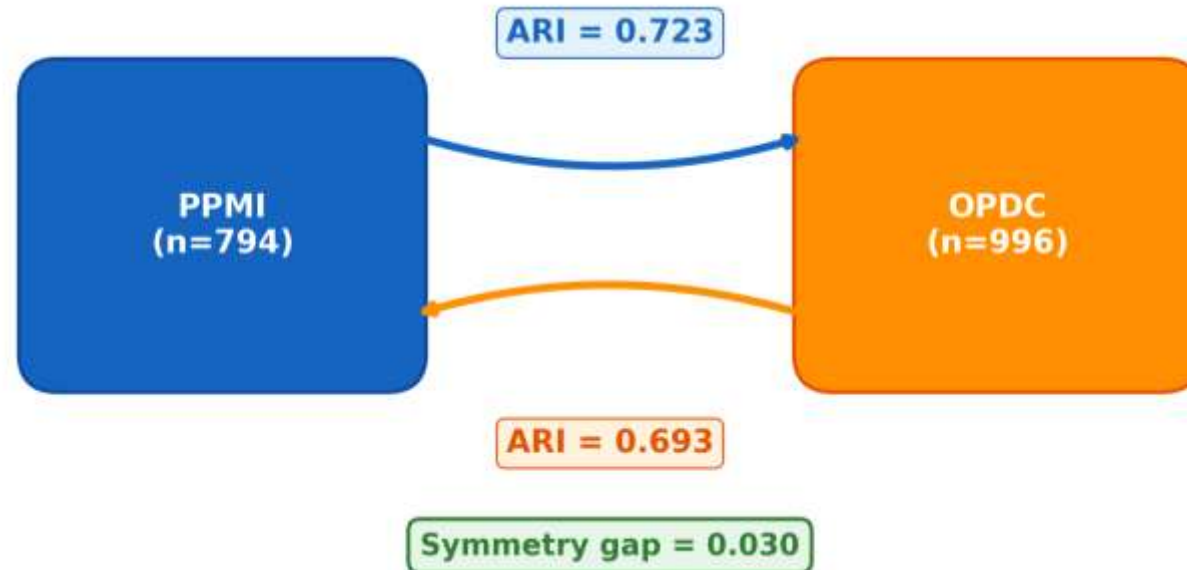
Harmonisation	ARI (500-boot)	Silhouette	Symmetry gap
None	0.142	0.284	0.322
Pooled Z	0.142	0.284	0.322
ComBat-EB	0.797	0.327	0.156
CORAL	<u>0.756</u>	<u>0.324</u>	0.010

➤ **CORAL provides the most reproducible clustering across cohorts**

Clustering & Replication

Why Symmetry Matters?

- Replication is Bidirectional:

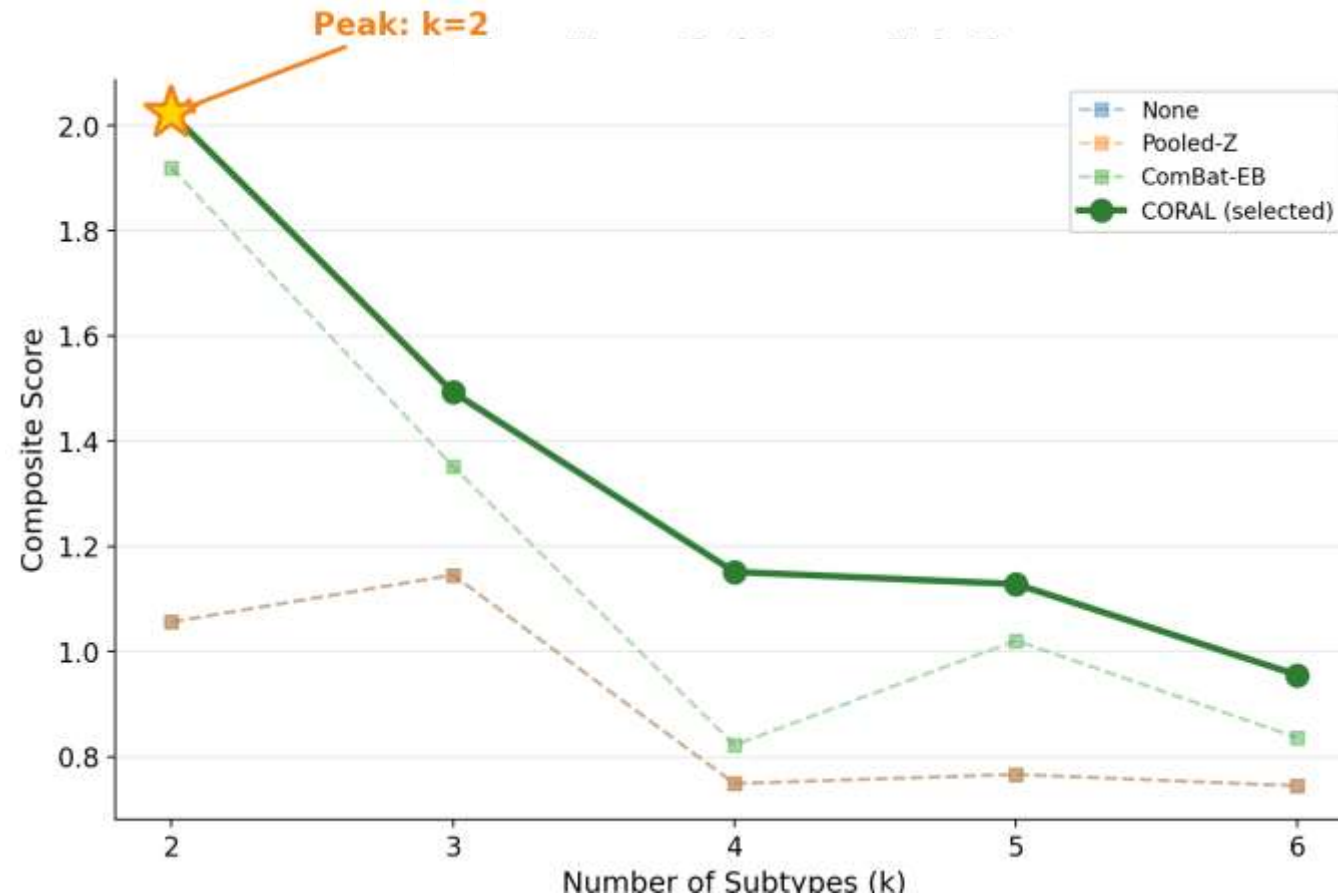


- **Subtypes are not cohort-specific, they transfer both ways.**

Clustering & Replication

Optimal Number of Subtypes

How Many Subtypes Exist?



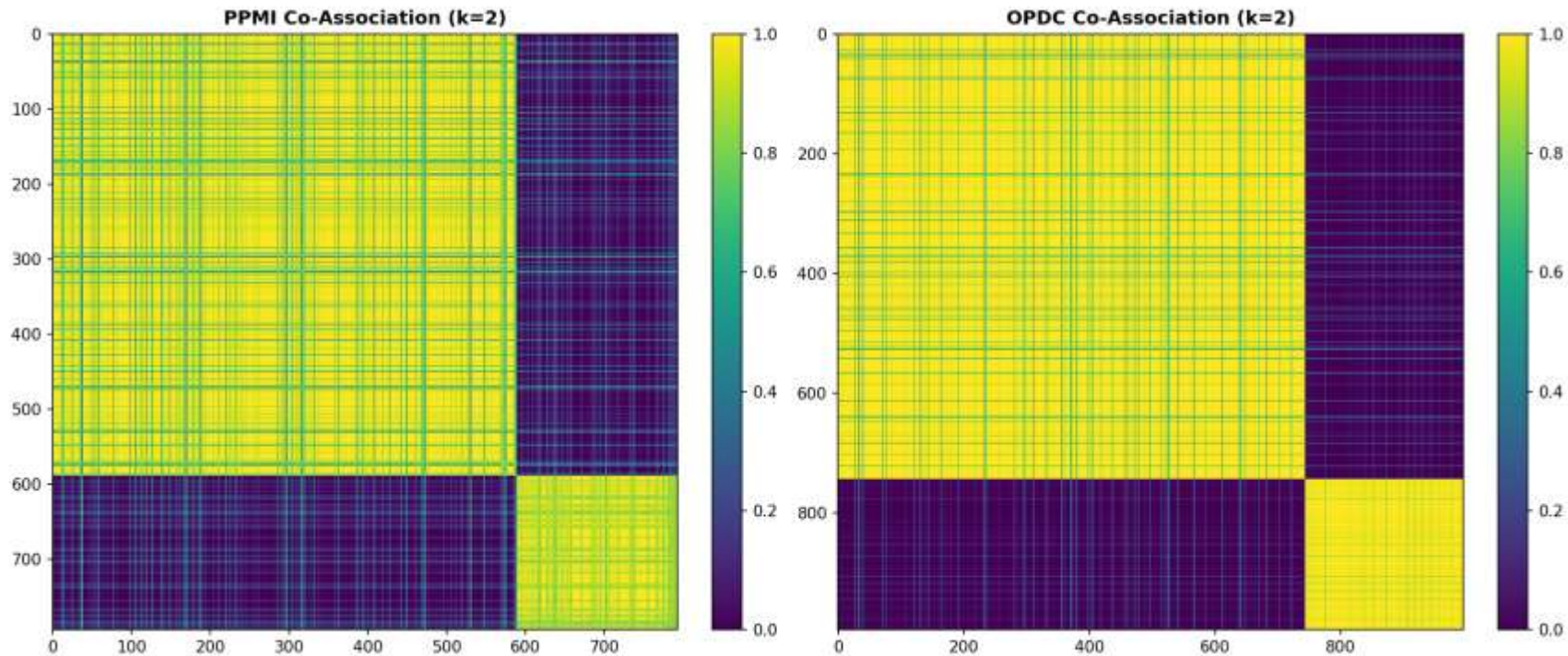
Peak at $k = 2$

➤ A stable bimodal structure emerges in both cohorts.

Stability

Consensus Clustering Stability

- Robustness of Patient Grouping:



PAC PPMI = 0.339
PAC OPDC = 0.132

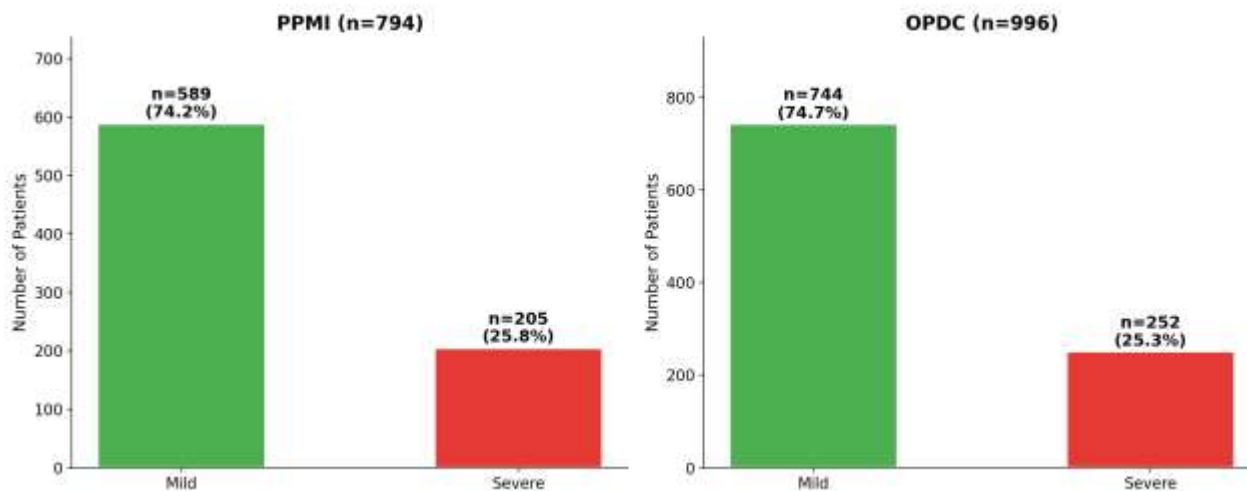
➤ **Moderate stability reflects genuine boundary uncertainty in a continuous phenotype**

Biological Meaning

Demographic Characterization

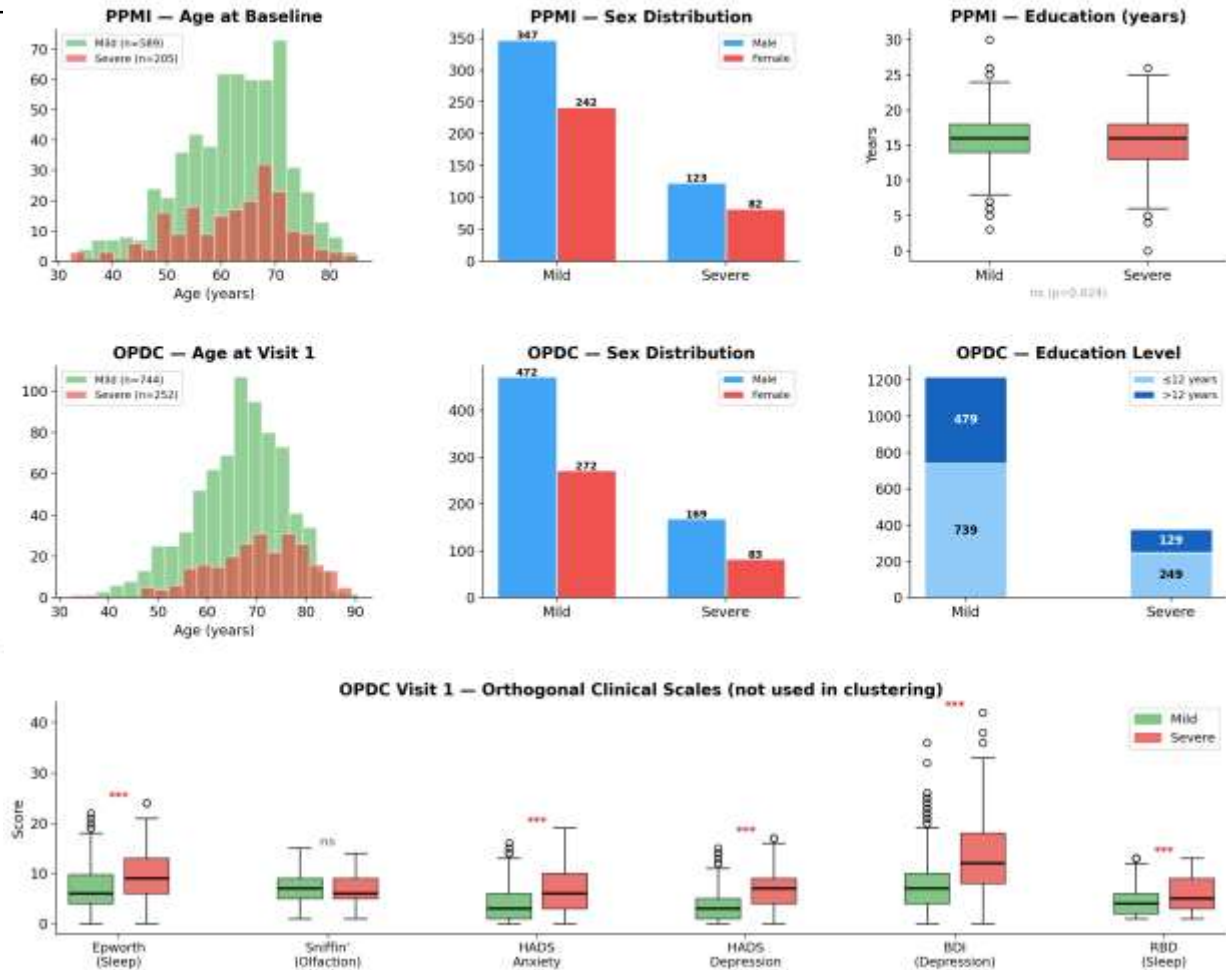
Two Clinically Distinct Profiles

Number of Patients in Each Subtype



Both cohorts show a majority in the Mild subtype, consistent with a severity gradient.

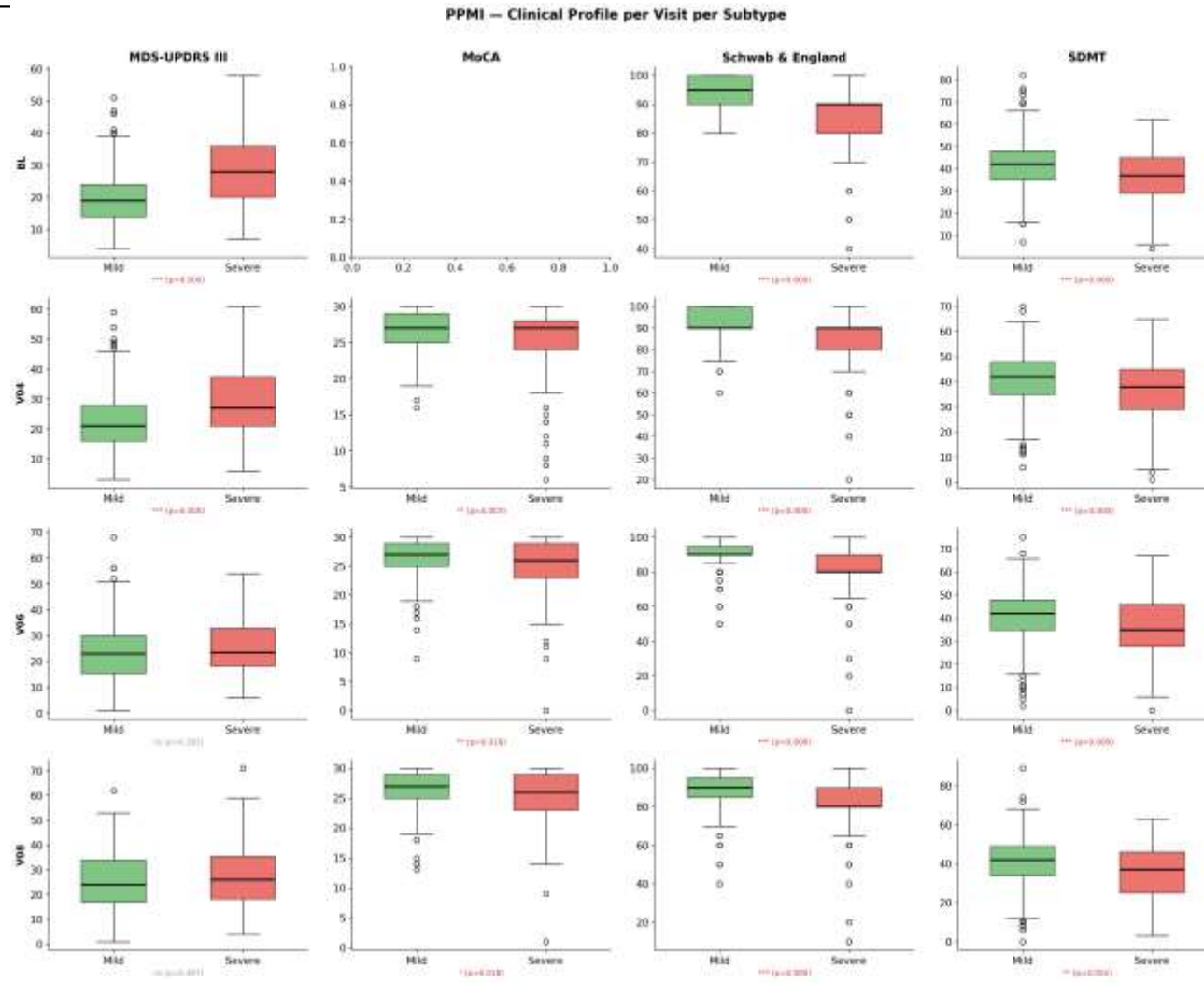
Demographic & Orthogonal Clinical Characterization by Subtype



Biological Meaning

Clinical Characterization

Two Clinically Distinct Profiles: **PPMI Clinical Profiles:**

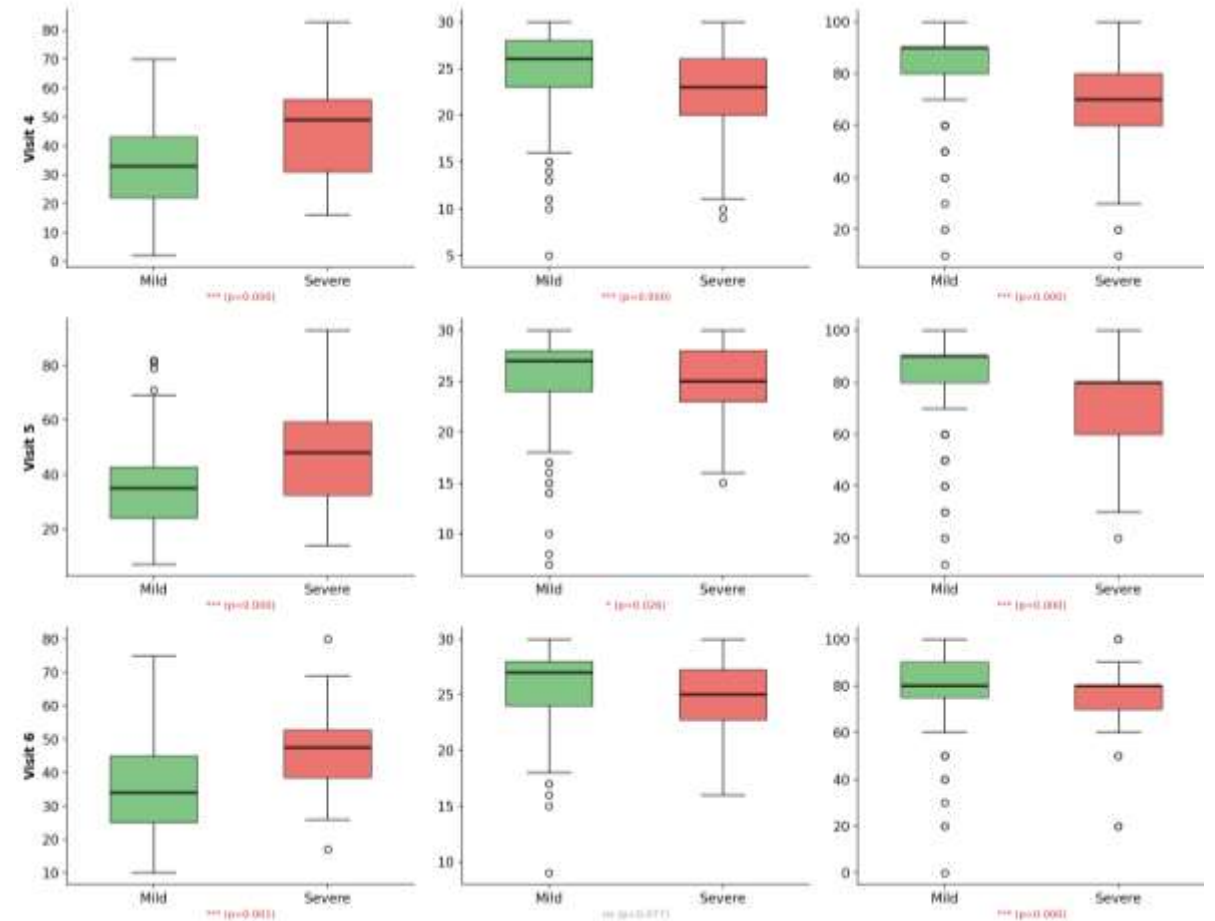
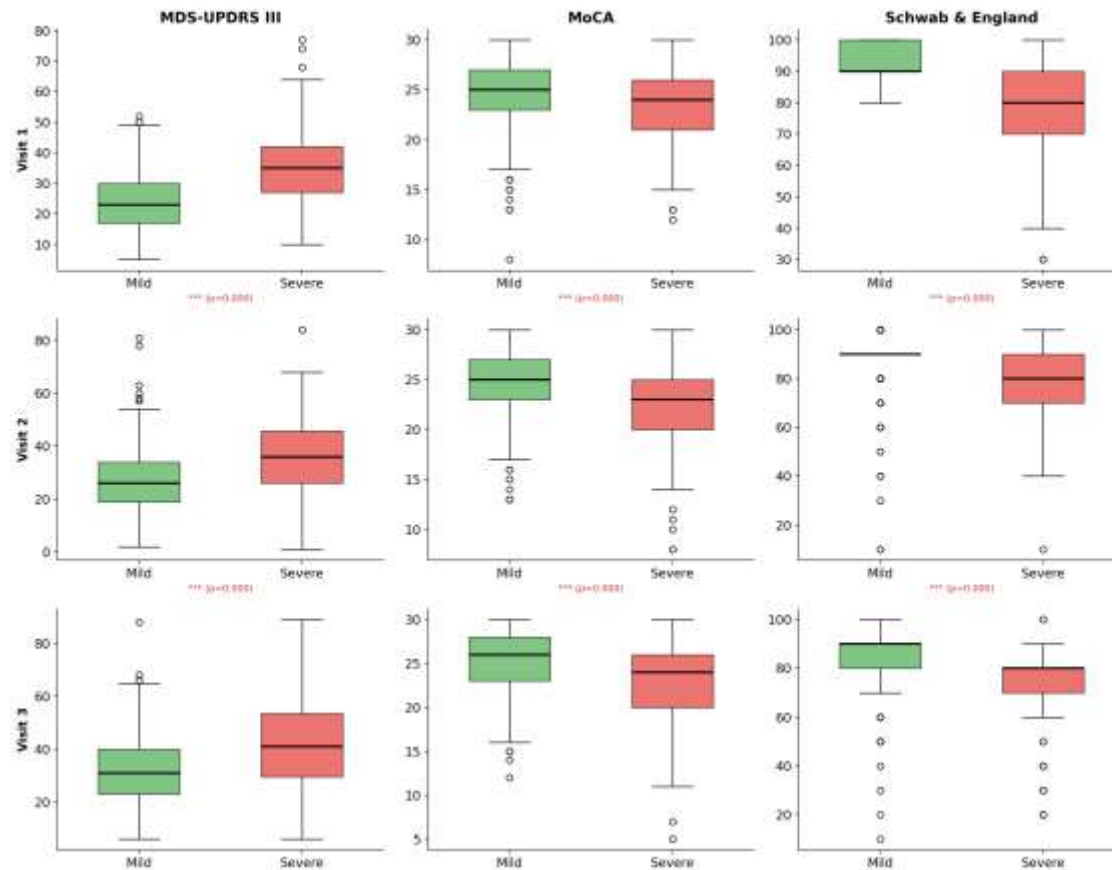


Biological Meaning

Clinical Characterization

Two Clinically Distinct Profiles: **OPDC Clinical Profiles** :

OPDC — Clinical Profile per Visit per Subtype



➤ Subtypes differ longitudinally on biomarkers used for clustering.

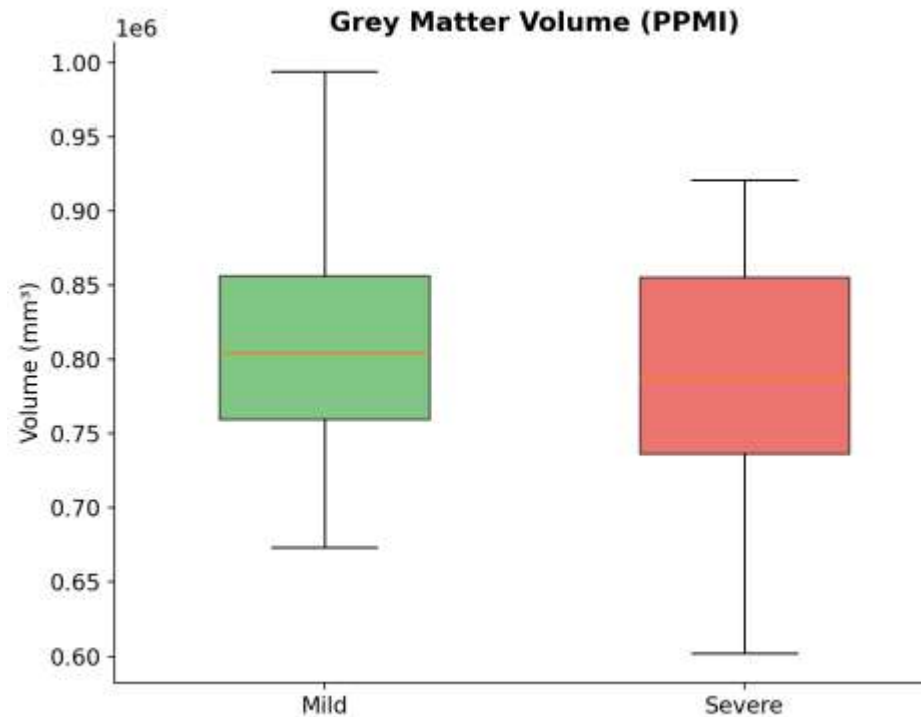
Biological Meaning

External Biomarker Validation

Independent Biomarkers Confirm the Subtypes: **Imaging**

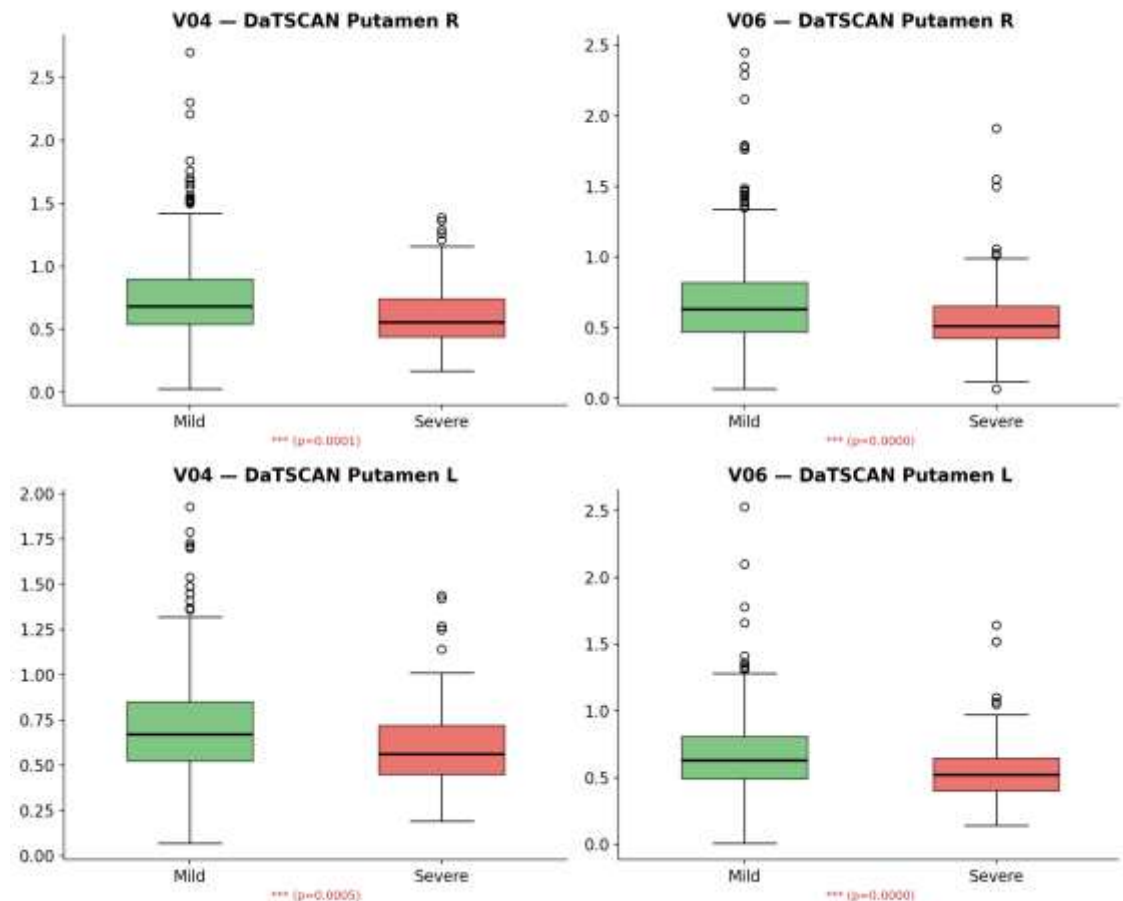
Biomarkers (DDMT)

Independent Biomarkers Confirm the Subtypes



Subtypes differ on biomarkers not used for clustering.

PPMI Imaging Biomarkers by Subtype Across Visits

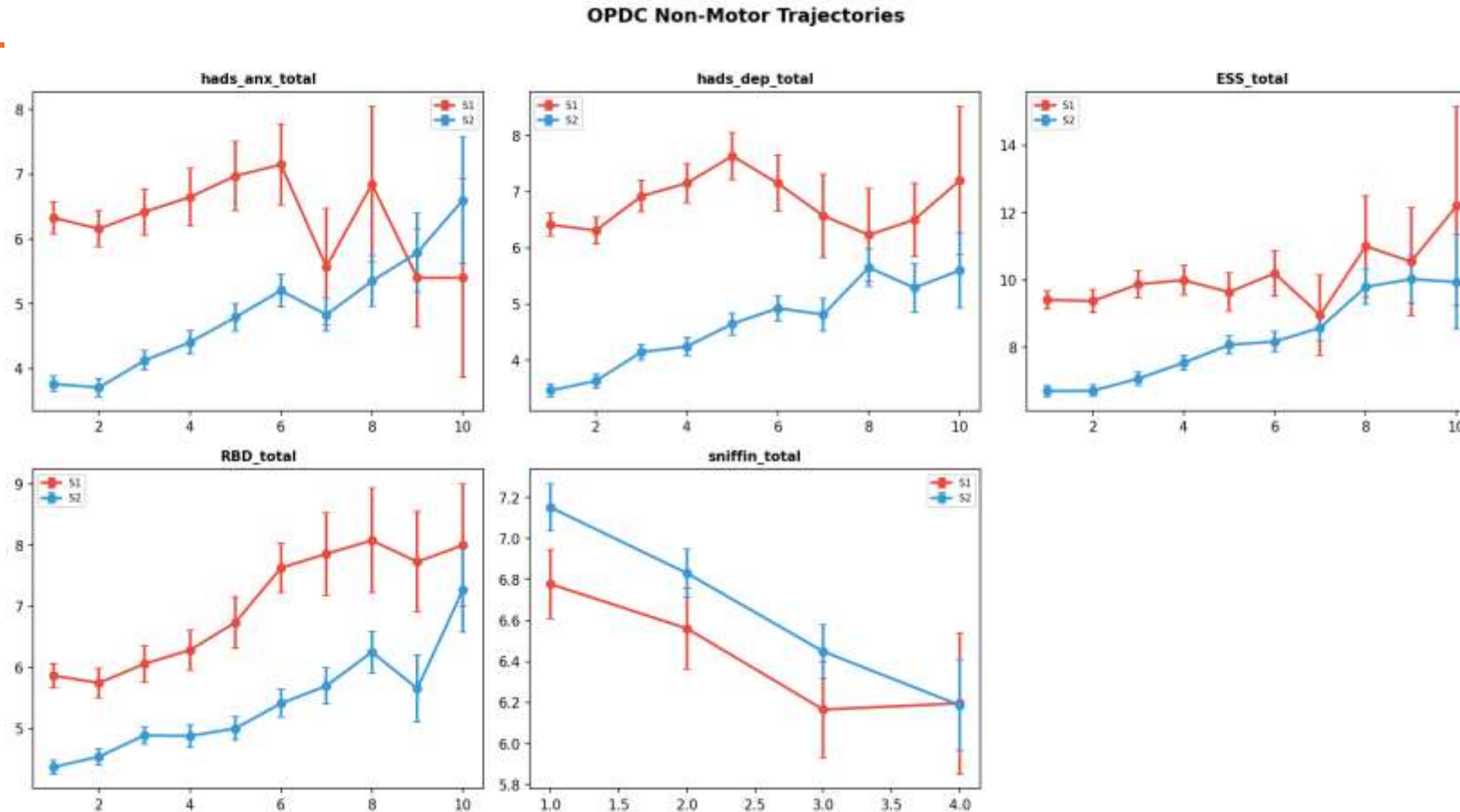


DaTSCAN confirms lower dopaminergic uptake in Severe subtype. No imaging for OPDC.

Biological Meaning

External Biomarker Validation

Independent Non-motor scales Confirm the Subtypes: Non-Motor Scales

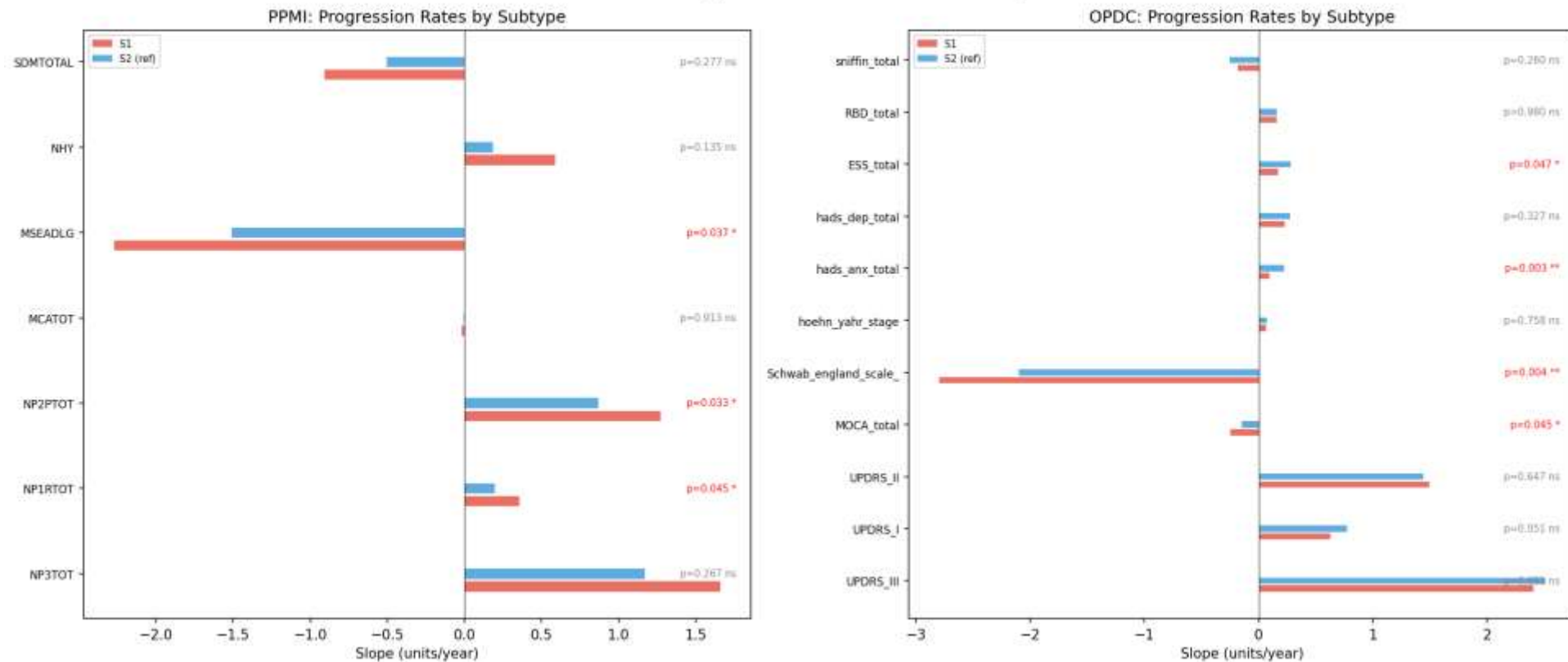


➤ Subtypes differ on biomarkers not used for clustering.

Disease Progression Longitudinal Trajectories

Different Disease Progression Rates:

LMM: Baseline Subtype Predicts Differential Progression



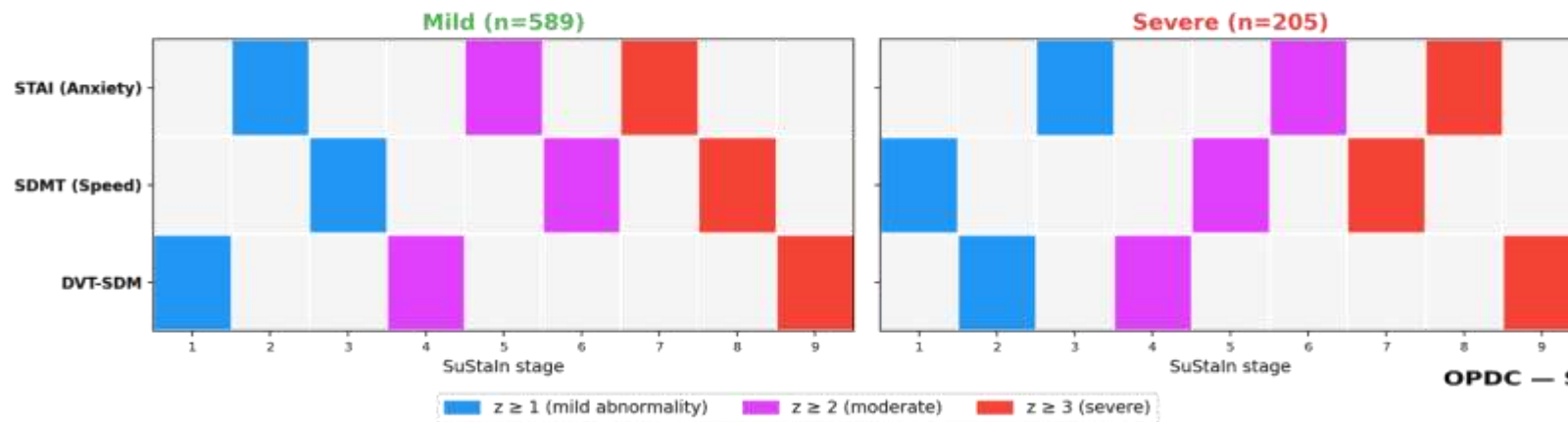
➤ **The severe subtype progresses faster in both cohorts.**

Disease Progression

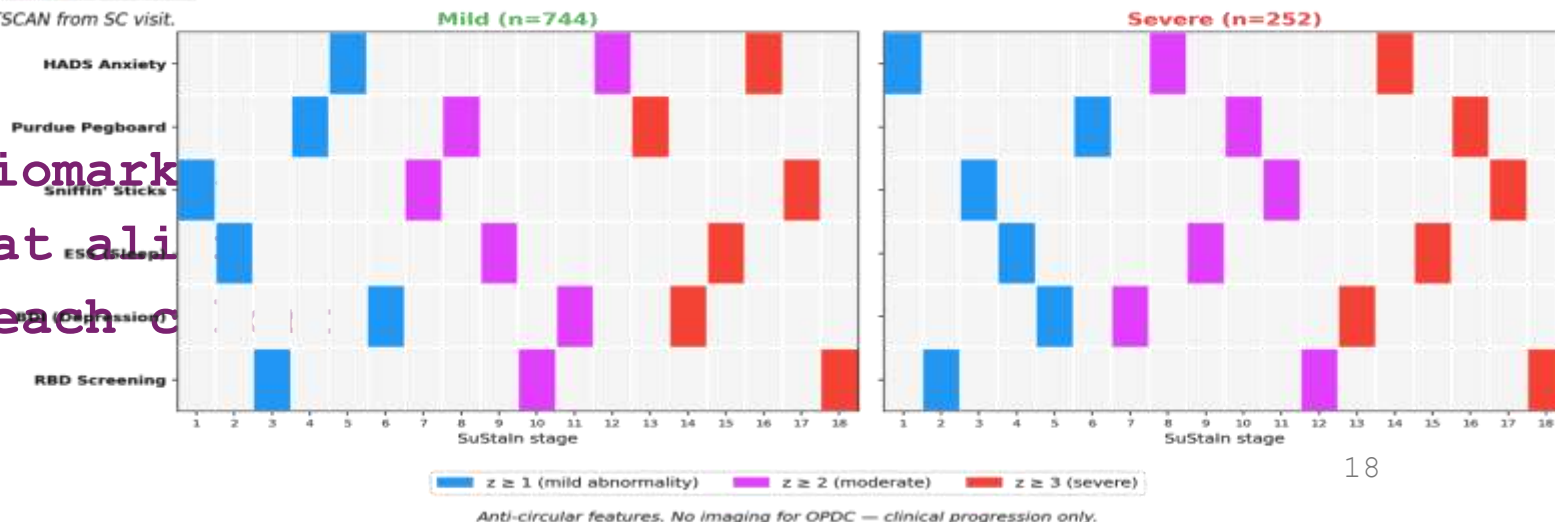
SuStain Disease Staging

Temporal Disease Structure:

PPMI — SuStain Progression per Subtype



OPDC — SuStain Progression per Subtype



SuStaIn reveals coherent biomarker deterioration sequences that align with the severity axis in each cohort.

Final Interpretation

What We Actually Proved?

Key Finding:

Parkinson's subtypes replicate across independent cohorts

- not caused by recruitment differences
- not caused by measurement differences
- reflects a shared disease severity axis

Final Interpretation

Take-Home Message

1

We did not
propose new
subtypes.

2

We validated
that the same
biological
structure
exists across
cohorts.

3

**This supports
subtype-based
precision
medicine in
Parkinson's
disease.**

Next steps

1. Feature Expansion

Integrate OPDC imaging assessments to evaluate whether more than two subtypes ($k \geq 3$) can replicate.

2. Third-Cohort Transfer

Apply the consensus subtype model to external cohorts (e.g., Tracking Parkinson's, ICICLE-PD) to test generalisability.

3. Purpose-Driven Assessment

Evaluate whether baseline subtype membership predicts clinically relevant outcomes, including treatment response.

4. Prospective Longitudinal Modelling

Use joint longitudinal models to formally test subtype $\times$ time interactions and progression dynamics.

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